Finance Engineer Cup Challenge Instructions

Build a reusable workflow to check RelayCloud customer contracts and usage against billing, explain findings and recommend actions. All companies, people and records are fictional. Use any tools or interface; no live system access is needed.

REQUIRED SCOPE: October 1-31, 2026 service and its effect on October books. Close cutoff: 2026-11-11T18:00:00-07:00. This is business day 8 using Monday-Friday and ignoring public holidays. The calendar definition only establishes the cutoff; no business-day calculation is required.

Use adjacent-period records where needed to explain October. Flag later-service issues as follow-ups; do not reconcile November or rebuild future contract history. Compare expected and actual customer billing with platform fees and usage shown separately. An annual prepaid platform fee is not billed again in October, but October earned revenue still exists.

For advance subscriptions, reconcile the invoice billing periods covering October and identify any additional October-service charges. You may instead show explicit day-based allocations to October, provided your basis is clear. Neither a full crossing-period invoice nor its billing allocation automatically equals October recognized revenue. Both well-supported presentation approaches are accepted.

Show errors, valid explanations and uncertainty with customer, affected period, supportable amounts, evidence and recommended next step. Customer complaints are not required to identify a finding. Results can be inside your build or exported; a separately formatted exception register is not required.

Submit the working build with run instructions, findings and supported October journal entries. A 3-5 minute walkthrough should demonstrate one important finding, one judgment call and a rerun; the artifact carries remaining detail. AI is welcome, but extracted terms and calculations must be reviewable. No future revenue schedule or system-record export is required; recommend updates instead.

COMPANY RULES: One USD entity/book, one platform subscription and one workflow-run usage SKU. Tax, FX, collections, cash settlement, processing fees, marketplace sales, breakage and complex multi-obligation arrangements are excluded.

PAYMENT AND DATE DISPLAY: Order forms use Net 30 payment terms. Collections, aging and settlement remain outside the exercise; unpaid invoices alone do not imply a required adjustment. Visible service ends are inclusive; the equivalent end-exclusive boundary is stated in the commercial terms and retained in system exports.

SOURCE PRECEDENCE: Executed order forms and amendments determine commercial terms. An amendment changes only named terms; unchanged terms persist. Register supersedes_document_id denotes partial precedence, not cancellation of the original order. START_HERE defines exercise accounting policy; messages establish approvals and context.

USAGE: Aggregate unique active records across a customer's verified product accounts before applying its calendar-month allowance. No rollover or tiers. Daily records are additive across product accounts but versions/duplicates must not be double counted. Explicit zero-use days remain records. record_status and replaces_record_id describe source corrections; there are no superseded versions in this release.

Partial first-month allowance: monthly allowance x active calendar days / days in month, rounded once to whole runs half up. For an allowance amendment, weight each allowance by applicable calendar days and round once. Charges equal max(0, eligible units minus allowance) x signed overage rate, rounded once to USD cents half up. Prices are negotiated net prices, with no additional invoice discounts.

UNVERIFIED USAGE: Reconcile known mapped activity separately and isolate unverified usage. Do not force a mapping, assume the remaining customer owns a workspace, or treat missing attribution as zero usage. Document conditional exposure and request Product Operations evidence. An unapproved concession request alone does not authorize a credit or revenue reduction.

BILLING: Platform invoices are issued in advance on contract billing anchors, either monthly or annually. Usage is always calendar-month based and separately invoiced on the first day of the next month. An absent usage invoice can be valid when no overage exists. Include invoices issued by cutoff; November 15 invoices have not yet been issued. September platform invoices are settled; October/November invoices remain unpaid.

ACCOUNTING: Recognize platform consideration by actual service days / 365, independent of invoice frequency. Preserve the original base schedule when adding an upgrade increment. Round each monthly recognition component once to cents, with final-period rounding allocated to the last period. The signed amendment defines incremental consideration and billing conventions.

Platform invoice: Dr 1100 AR / Cr 2200 deferred platform revenue. Recognition: Dr 2200 / Cr 4000 platform revenue. Any customer platform net debit in 2200 is reclassified Dr 1160 platform contract asset / Cr 2200. Reverse any prior reclass before recalculating. The supplied October GL ALREADY contains its baseline reclass. Propose only incremental changes after correcting recognition. Do not net different customers or platform assets against usage.

Usage accounting: Dr 1150 unbilled usage receivable / Cr 4010 usage revenue in the service

month. The next-month invoice is Dr 1100 / Cr 1150, not revenue again. September 30 unbilled balances are released by October 1 invoices; November 1 invoices release October accruals only to the amounts actually billed. Residuals may remain until supported billing corrections.

CREDITS: Use issued notes only; void notes have no current credit effect. Retain original service-period allocation. Approved reductions of October consideration affect October where not already reflected. A November credit can reduce AR and reverse the corresponding release of unbilled usage without reducing revenue a second time.

APPROVAL AND DATES: Recommend corrections for Controller approval. Finalized invoices are not silently edited: propose supplemental invoices or linked credit notes. Proposed billing documents use November issue dates; supported October-service accounting corrections use the October posting period. You need not create live records.

TIMING: Source activity, ingestion timestamp, invoice issue date, service period, GL posting period and GL Created At are distinct. October entries can be created during November close. Determine evidence availability using the cutoff, not an assumption that October entries were all created October 31.

FORMATS: UTF-8 comma-delimited CSV, one header, blank means missing/not applicable rather than zero. IDs are strings. ISO dates; normalized ISO timestamps include America/Denver UTC offsets. All normalized service intervals are start-inclusive/end-exclusive. These reports normalize source dates/units; they are not raw API payloads.

MONEY: Stripe amount/subtotal/total/discount fields are integer USD cents; unit_amount_decimal is decimal cents per unit. NetSuite amounts are decimal USD. Use exact decimal arithmetic, round half up. Debit/Credit fields are unsigned; signed movement = Debit minus Credit. USD is the sole currency.

EXPORT PROVENANCE: Salesforce configured Details Only account-report CSV; cross-system identifiers are custom fields. Stripe invoice/line custom reports based on Sigma structures; credit files are flattened API extracts with service-period enrichment from invoice lines or approval evidence. NetSuite configured posting/saved-search and ARM plan-detail extracts. Headers are configured company selections, not claimed universal defaults.

RECONCILIATION POPULATION: 30 customers, 32 product accounts, 30 executed order forms, 2 amendments; 1,924 unique source daily usage records across 61 dates. Physical export row count can differ and must be reconciled. 120 customer opening-account rows; 362 existing plan-period rows. Independent usage_batch_controls are source counts/totals, not a commercial-billability answer.

CONTROLS: September 1-14 source batches have 30 records/day; September 15-October 31 have

32. Opening balances include accounts 1100, 1150, 1160 and 2200 for every customer, including zeros. September recognition journal IDs in plans are historical references outside the GL period; October recognized plan rows must join included postings. Future plan rows are context.

EXTRACT COMPLETENESS: These are all issued invoices and credit notes for the supplied customer population through cutoff, and all relevant-account GL posting lines in the stated periods. No pagination or file truncation is intended. Source report row counts: contract_register.csv = 32, netsuite_gl_detail.csv = 340, netsuite_opening_balances.csv = 120, netsuite_revenue_plan_detail.csv = 362, product_account_mapping.csv = 32, salesforce_accounts.csv = 30, stripe_credit_note_lines.csv = 2, stripe_credit_notes.csv = 2, stripe_invoice_lines.csv = 124, stripe_invoices.csv = 124. Compare these counts to the supplied files; they do not establish that a commercially required invoice was actually issued.

PRODUCTS: prod_platform = platform subscription; prod_workflow_run = workflow-run overage. Price IDs are customer-specific. Product-account labels are display text, not a substitute for verified mappings.

FILE DICTIONARY:

contract_register.csv: One document indexed by document_id. Filename locates PDF. contract_id groups order/amendment. signed_at, effective dates and billing_loaded_at are distinct; blank billing load means not recorded as loaded. approval_reference links a message where applicable.

Columns: document_id, contract_id, salesforce_account_id, document_type, document_filename, signed_at, effective_from, effective_to, supersedes_document_id, billing_loaded_at, approval_reference

netsuite_gl_detail.csv: Internal ID + Line ID identifies posting line; all legs supplied. Date, Posting Period and Created At differ. External Source ID joins invoice/credit, Revenue Plan ID joins recognition plan where applicable. Memo is descriptive. October and relevant November activity supplied through cutoff.

Columns: Internal ID, Line ID, Transaction Type, Document Number, Date, Posting Period, Account Number, Customer Internal ID, Debit, Credit, Memo, External Source ID, Revenue Plan ID, Created At

netsuite_opening_balances.csv: One Customer Internal ID + Account Number at September 30. Debit Balance/Credit Balance are USD; only one nonzero. Source Reference links supported closing workpaper. All four balance-sheet accounts are explicit per customer.

Columns: Customer Internal ID, Account Number, Account Name, As Of Date, Debit Balance, Credit Balance, Source Reference

netsuite_revenue_plan_detail.csv: Plan + element + posting period + source document identifies recognition segment. Recognition Start/End Exclusive are service dates. Plan Type Actual is not a forecast. Planned Amount and Recognized Amount are USD; Journal Internal ID references recognized postings. Plan Updated At is source configuration timestamp.

Columns: Revenue Plan ID, Revenue Element ID, Customer Internal ID, Source Document ID, Plan Type, Recognition Start, Recognition End Exclusive, Posting Period, Planned Amount, Recognized Amount, Journal Internal ID, Plan Updated At

product_account_mapping.csv: One product account. salesforce_account_id can be unverified/blank. Effective dates are date intervals; effective_to blank means no known end. mapping_updated_at records registry update.

Columns: product_account_id, salesforce_account_id, account_label, effective_from, effective_to, mapping_updated_at

product_usage.csv: One record/version exported from metering. record_id identifies source activity and is the deduplication key. units is integer runs. ingested_at records availability. batch_id joins daily controls. Exact repeated source IDs are not new activity.

Columns: record_id, product_account_id, sku, service_date, units, ingested_at, record_status, replaces_record_id, batch_id

salesforce_accounts.csv: One account per AccountId. Custom StripeCustomerId__c and NetSuiteCustomerInternalId__c join source customers. ParentId is nullable; owner and billing country are descriptive.

Columns: AccountId, AccountName, ParentId, BillingCountry, StripeCustomerId__c, NetSuiteCustomerInternalId__c, AccountOwner

stripe_credit_note_lines.csv: One credit line by id. credit_note_id joins note; invoice_line_item_id joins original line. amount is positive cents. quantity nullable. Service dates are internal enrichment, with service_period_source identifying provenance.

Columns: id, credit_note_id, invoice_line_item_id, description, amount, quantity, service_start, service_end_exclusive, service_period_source

stripe_credit_notes.csv: One note by id, joined to invoice_id/customer_id. amount is

positive credit cents. status issued/void determines effect; voided_at is nullable. reason is a source reason code.

Columns: id, number, invoice_id, customer_id, status, created, voided_at, currency, amount, reason

stripe_invoice_lines.csv: One line by id. invoice_id joins header. product_id/price_id identify product and customer price. quantity is units (1 for platform). amount is before separate discount_amount. period_start/end are normalized exclusive service boundaries; proration is boolean.

Columns: id, invoice_id, description, product_id, price_id, quantity, unit_amount_decimal, amount, discount_amount, period_start, period_end, proration

stripe_invoices.csv: One header by id. customer_id joins Salesforce custom ID. created/finalized_at are timestamps, status is open/paid/etc. subtotal/total/total_discount_amount are cents. total = subtotal minus discount; taxes/fees/customer-balance effects excluded.

Columns: id, number, customer_id, status, created, finalized_at, currency, subtotal, total_discount_amount, total, source_system

usage_batch_controls.csv: One service date. distinct_active_record_count and active_units_total come from source before export. extracted_at is the cutoff snapshot. Reconcile after deduplication, not after dropping unmatched customers.

Columns: service_date, batch_id, distinct_active_record_count, active_units_total, extracted_at